

Project 4 — Quantitative Structural Comparison of Lysozyme Structures

Research Question

How structurally similar is hen egg-white lysozyme when it is free versus when it is bound to an antibody?

Structures Compared

Feature	1AKI	1FDL
Protein	Hen egg-white lysozyme	Hen egg-white lysozyme in Fab D1.3 complex
Chain	A	C (author chain Y)
Residues	1–129	1–129
Method	X-ray diffraction	X-ray diffraction
Resolution	1.50 Å	2.50 Å

Alignment Method

RCSB Protein Data Bank pairwise structure alignment using TM-align. Reference: 1AKI Chain A, residues 1–129. Second structure: 1FDL Chain C (author chain Y), residues 1–129.

Quantitative Results

Measure	Result
RMSD	0.37 Å
TM-score	0.99
Sequence identity	100%
Aligned residues	129 / 129
Length compared	129 / 129

Interpretation

The RMSD of 0.37 Å is very small and the TM-score of 0.99 indicates extremely high global structural similarity. All 129 residues aligned and sequence identity was 100%. These results indicate that the overall 3D fold is highly conserved between the free and antibody-bound lysozyme structures.

Biological Significance

The comparison supports the earlier qualitative analysis: antibody binding does not appear to cause a major change in the global fold of hen egg-white lysozyme.

Important Caveat

A very low global RMSD does not mean every residue is unchanged. Local conformational differences can still occur, particularly around an interaction interface. In 1FDL, antibody contacts with lysozyme include residues 18–27 and 117–125. Residue-level comparison would be needed to investigate local effects.

What I Learned

This project moved from qualitative visualization to quantitative structural analysis. I learned what RMSD and TM-score represent, how sequence identity relates to structural comparison, and why global metrics need biological context.

Limitations

The analysis focused on global similarity. It did not calculate residue-level distance changes, backbone-angle differences, interface energetics or detailed conformational changes around the binding site.

Next Step

A useful follow-up would be to compare the antibody-contacting residues individually between 1AKI and 1FDL to determine whether small local changes exist despite the very high global similarity.

Sources

RCSB Protein Data Bank • RCSB Pairwise Structure Alignment • TM-align